Individual Student Attention Detection in Face-to-Face Classrooms Using Multimodal Facial and Wearable Data

Jiaqi Liu
School of Science and Technology
Hong Kong Metropolitan University
Hong Kong, China
s1304012@live.hkmu.edu.hk

Kwok Tai Chui
School of Science and Technology
Hong Kong Metropolitan University
Hong Kong, China
jktchui@hkmu.edu.hk

Lap-Kei Lee
School of Science and Technology
Hong Kong Metropolitan University
Hong Kong, China
lklee@hkmu.edu.hk

Kwan Keung Ng
Ming-Ai (London) Institute
London, United Kingdom
prof.stevenng@gmail.com

Naraphorn Paoprasert
Department of Industrial Engineering
Faculty of Engineering
Kasetsart University
Bangkok, Thailand
naraphorn.p@ku.th

Mingbo Zhao
School of Information Science and Technology
Donghua University
Shanghai, China
mzhao4@dhu.edu.cn

***Abstract*—Accurately assessing individual student attention in real-world classrooms remains a significant challenge due to the limitations of subjective teacher observation and the inherent complexity of human behaviour. This work presents a robust multimodal framework that integrates synchronised facial video and wearable sensor data to objectively and continuously estimate student attention in face-to-face classroom settings. Our system combines visual, physiological, and motion cues for fine-grained attention prediction by leveraging a deep neural architecture with modality-specific encoders and advanced fusion techniques. Evaluations on the large-scale DIPSEER dataset demonstrate that our approach outperforms single modality baselines and achieves high precision in regression and classification tasks (root mean squared error of 0.87 and one-off accuracy of 76.3%). The results highlighted the effectiveness and generalisability of multimodal fusion for attention detection, providing a scalable and reliable solution for automated classroom analytics.**

***Keywords*—deep learning, education, facial analysis, multimodal learning, student attention detection, wearable sensors.**

I. Introduction

Student attention plays a critical role in effective classroom learning. The ability of students to remain focused during lessons directly impacts their understanding, engagement, and academic performance [1-3]. In traditional face-to-face classrooms, teachers often rely on subjective observation and experience to assess student attention, which can be imprecise and labour-intensive, especially in large or dynamic settings [4]. As educational environments become increasingly diverse and technology-enhanced, there is a growing need for objective, scalable, and automated methods to monitor and evaluate student attention in real-time [5].

Recent advances in sensing technologies and machine learning have enabled the development of intelligent educational technologies capable of capturing multifaceted aspects of student behaviour [6]. Integrating visual data, such as facial expressions and head pose, with physiological and behavioural signals from wearable devices offers a promising pathway for comprehensive student attention analysis. This multimodal approach allows for the fusion of observable external cues and internal physiological states, providing a more robust and reliable measure of attention than single-modality systems [7,8].

Despite these advancements, most existing research has focused on either simulated environments or relied on a single data modality, limiting the ecological validity and generalizability of findings. The unique challenges in real-world classroom settings, such as natural variations in student posture, lighting conditions, and subtle behavioural cues, necessitate more sophisticated and adaptable recognition systems [9-11]. Furthermore, the combination of facial video and wearable sensor data in classroom environments remains underexplored, particularly in modelling individual attention dynamics across diverse learners.

This study proposes a multimodal framework for automatic student attention recognition in physical classroom settings. It leverages synchronised facial video and smartwatch sensor data to capture visual and physiological indicators of attention. A deep learning-based fusion model is designed to integrate features from both modalities to predict attention levels on a continuous scale. Through extensive experiments on a benchmark dataset collected from real classroom sessions, it is demonstrated that the proposed multimodal method outperforms single modality baselines and provides a reliable, objective assessment of student attention.

This research advances the field of automated classroom engagement analysis by integrating multiple complementary data modalities, including facial video, physiological signals, and motion sensor data, to enable a more comprehensive and fine-grained understanding of student attention and emotion. Unlike prior approaches that rely solely on visual cues or behavioural logs, the proposed method captures both externally observable behaviours and internal physiological states, thus providing a richer representation of engagement that is less susceptible to ambiguity or occlusion in a single modality. This multimodal fusion approach enhances the accuracy and robustness of engagement prediction and lays the groundwork for scalable, real-time systems capable of continuous monitoring in real classroom environments. Ultimately, the key research contribution is a new methodological foundation for intelligent learning analytics and supports the development of personalised, adaptive educational interventions grounded in a deeper understanding of student engagement [12].

2025 International Symposium on Educational Technology (ISET) | 979-8-3315-9550-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/ISET65607.2025.00017

2766-2144/25/$31.00 ©2025 IEEE
DOI 10.1109/ISET65607.2025.00017

II. Related Work

A substantial body of recent research has leveraged computer vision techniques to automatically assess student engagement in both online and physical classroom environments. These existing methods typically focused on extracting behavioural cues from facial features, gaze direction, and bodily actions.

Qi et al. [13] proposed a cascaded analysis network that integrated gaze estimation, facial expression recognition, and action recognition to classify and grade student engagement in online classes. The network hierarchically analyzed whether a learner is present and focused on the screen, analyses facial expressions to infer engagement, and, when the learner is not facing the screen, recognises actions such as writing or reading to assess continued engagement. It demonstrated that the engagement prediction accuracy could be improved over classical end-to-end network baselines, yet it primarily relies on vision-based signals obtained from camera data. While fusion across gaze, expressions, and actions enhanced robustness, the framework did not incorporate any physiological or inertial sensor data, potentially missing subtle engagement cues that were not visually observable. Likewise, a real-time attention monitoring system was designed using deep learning [14], where student activity and emotion are recognised from classroom images via the YOLOv5 object detector. The detector distinguished between high- and low-attention behaviours (e.g., raising hands, writing, using a phone) and classified facial emotions even when students wear masks. The pipeline included a tracking module (DeepSORT) for persistence in identity, and face-based attendance was monitored using conventional recognition algorithms. The approach offered real-time feedback and handled occlusions due to masks, yet the recognition is fundamentally limited to observable visual cues. Other relevant signals of internal cognitive or physiological engagement remain inaccessible in this framework.

Another work [15] took a different perspective by proposing a multimodal data-supported learning engagement analysis. Yet, the data modalities in their study were restricted to online platform behavioural logs (e.g., forum interactions and resource views), facial expression analysis, and discussion content. Their method employed machine learning to map behavioural indicators to engagement levels and used tools (such as FaceReader) for emotion extraction from video. The study demonstrated strong consistency between data-driven engagement estimations and self-reports, particularly for behavioural and cognitive dimensions. However, their notion of multimodality was limited to digital traces and visual emotion; no bio-signals or physical activity sensors were included, thus excluding a range of engagement-related physiological data.

In the context of physical classroom behaviour detection, Wang et al. [16] introduced the SCB-DETR framework, which utilized a multi-scale deformable transformer for robust detection of classroom behaviours such as writing, reading, raising hands, and discussion. Their model was trained on a custom-labelled dataset and achieved state-of-the-art detection under challenging conditions like occlusion or scale variance. While this method advances the robustness and accuracy of vision-based detection and can fine-grained action classification, it remains fundamentally dependent on video-based observable behaviours. It does not address internal cognitive or affective states beyond what is externally manifest.

The literature above demonstrates that vision-based techniques—whether through facial expression analysis, gaze tracking, or action recognition—have become the de facto standard for automated engagement assessment. They excel at unobtrusively monitoring students in real time, handling occlusions, and providing interpretable cues to instructors. Multimodal fusion within the visual domain (e.g., combining gaze and action) further enhances performance and reliability. However, these approaches share a notable limitation: their modality space is confined to what is observable through camera or platform logs. These frameworks do not consider signals that reflect physiological arousal, stress, or subtle cognitive engagement (such as heart rate variability or micro-movements captured by inertial sensors). As a result, engagement assessment is inherently restricted to externally visible features, which may not fully capture the complexity of student attention and affective states, particularly in scenarios where behavioural cues are ambiguous or masked.

In summary, while recent methods have demonstrated satisfactory performance in visual engagement assessment and robust performance in classroom settings, there remains a methodological gap in integrating non-visual modalities that can provide a more comprehensive and nuanced understanding of student engagement.

III. Methodology

A. Data Preprocessing

DIPSEER dataset [17], a large-scale, multimodal corpus specifically designed for in-person student attention recognition in real classroom environments, is chosen as a benchmark dataset to evaluate the performance of the proposed method. It comprises 54 students from three pre-existing groups, captured during diverse educational activities such as lectures, presentations, tests, and collaborative sessions. For each participant, two primary data streams were used: facial images and smartwatch sensor data. The facial data comprises high-resolution RGB images captured at a nominal rate of 10 frames per second (fps), with each image representing a static snapshot of the student's face. Due to hardware limitations and real-world recording conditions, the number of facial images per second may fall below ten, resulting in occasional frame loss during acquisition.

Smartwatch sensor data were recorded continuously, including heart rate at 1 Hz and inertial measurements (linear acceleration and gyroscope) at 100 Hz along three axes. All sensor data is stored in JSON format and is timestamped for synchronisation with the facial images. Only these two data modalities, facial images and smartwatch sensor streams, were utilised in this study.

The dataset is hierarchically structured for data organisation by group, experiment, and subject. The folder of each subject contains subfolders for the facial image sequence and the synchronised sensor data. During preprocessing, all data streams were first temporally aligned using their timestamps, ensuring accurate correspondence between each facial image (or group of images in a time segment) and the relevant sensor readings within the same interval. Data were then segmented into fixed-length, non-overlapping intervals (typically 1 to 10 seconds), each serving as a sample for supervised learning. The data preprocessing pipeline is shown in Fig. 1.

Segments with missing or corrupted facial images or sensor dropouts were identified and excluded to ensure data reliability. For the facial data, each image within a segment was processed with a face detection and alignment pipeline, followed by extraction of facial landmarks and head pose features; the resulting frame-level features were then aggregated over the segment using statistical descriptors such as mean and variance to account for variable frame counts. Raw accelerometer, gyroscope, and heart rate signals were resampled and normalised for the smartwatch data, and time- and frequency-domain features were computed for each interval. All extracted features were standardised across the dataset to facilitate multimodal fusion.

Attention labels, provided by multiple expert raters and self-evaluations on a five-point scale, were temporally aligned to each segment based on consensus or averaging. Through this rigorous preprocessing workflow, the final dataset consists of high-quality, temporally aligned, and feature-rich samples, each representing a short time interval with robust multimodal features and corresponding attention labels, suitable for downstream automatic attention modelling.

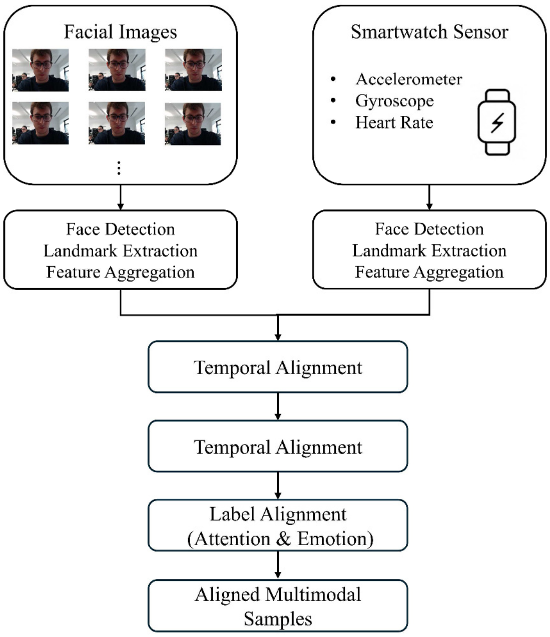


Fig. 1. Data Preprocessing Pipeline for Multimodal Attention Analysis.

B. Model Construction

Following data preprocessing and temporal alignment, we designed a robust multimodal neural architecture to jointly process synchronised visual and smartwatch sensor data for per-second attention and emotion analysis. To ensure the model's generalizability and resilience against overfitting, standard regularisation techniques, including dropout, batch normalisation, and layer normalisation, are applied throughout the architecture to improve generalisation and training stability as shown in Fig. 2.

1) Multimodal Feature Encoding: For each one-second window, the model receives three types of aligned input:

a) Facial Images: A dedicated visual encoder processes all facial images captured within the same second. This encoder utilises an EfficientNet-B0 backbone, initialised with pretrained weights [18]. The backbone incorporates intrinsic dropout to improve regularisation at the feature extraction stage. Each image is independently passed through the backbone, generating a set of high-level feature vectors.

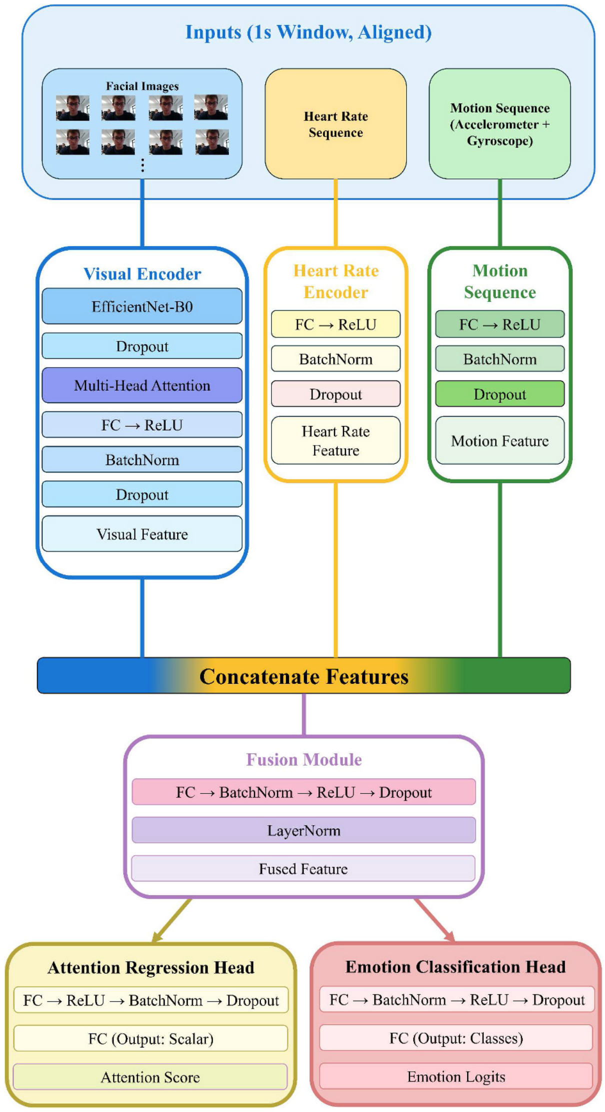


Fig. 2. The Proposed Multimodal Attention and Emotion Recognition Model.

b) Heart Rate Encoder: A feed-forward block encodes the per-second heart rate value. This consists of a linear transformation, ReLU activation, BatchNorm, Dropout, and a final linear layer to project the heart rate into the same feature space as the other modalities. Dropout is set at a moderate rate to regularise this low-dimensional signal.

c) Motion Encoder: The motion signal is a concatenation of six features: three-axis accelerometer and three-axis gyroscope statistics per second. This vector is encoded through another feed-forward network with a similar structure: linear layers, ReLU activation, BatchNorm, and a higher Dropout rate, which reflects the higher variability and potential noisiness of motion data. The output is a dense feature vector aligned in size with the visual and heart rate encodings.

2) Multimodal Feature Encoding: After independent encoding, the three modality-specific feature vectors are concatenated to form a comprehensive multimodal representation. This fused feature passes through a fusion block composed of fully connected layers, ReLU activations, batch normalisation, dropout, and layer normalisation. BatchNorm and Dropout are used to counteract overfitting and stabilise training, while LayerNorm mitigates internal covariate shift in the final fused representation. The resulting multimodal feature is fed into two parallel prediction heads:

a) Attention Regression Head: This head consists of multiple linear layers interleaved with ReLU activations, BatchNorm, and Dropout. An extra hidden layer and higher Dropout rates are applied, given the importance and difficulty of robust continuous attention score prediction. The output is a single scalar per sample, representing the predicted attention score for that second.

b) Emotion Classification Head: The emotion head is a multilayer perceptron with Dropout regularisation, mapping the fused feature to logits over the predefined set of emotion classes. This enables multi-task learning and supports comprehensive affective state analysis.

This architecture is fully end-to-end trainable and designed to leverage complementary temporal, visual, and physiological information for fine-grained attention and emotion analysis. The combination of advanced deep feature extraction, attention-based temporal aggregation, and strong regularisation ensures high expressiveness and robust generalisation across subjects.

C. Training and Validation

All model training and validation were performed on the pre-processed DIPSEER dataset. To ensure person-independent evaluation, the dataset was split at the subject level, resulting in 293 subjects for training, 98 for validation, and 98 for testing, for a total of 660 GB. After processing and filtering, the training, validation, and test sets contained 16,230, 5,351, and 5,322 samples, respectively. Each sample corresponds to a one-second window, containing up to 10 facial frames, synchronised heart rate, accelerometer, gyroscope readings, and corresponding expert and self-annotations.

During training, we used the AdamW optimiser and adopted a differentiated learning rate scheme, assigning different learning rates and weight decay values to different parts of the model. The visual encoder was trained with a lower learning rate (2×10^{-5}) due to its larger capacity and pretraining. In contrast, while the sensor encoders (heart rate and motion) were set to a moderate learning rate (5×10^{-5}), and the fusion and prediction heads were trained with a relatively higher learning rate (1×10^{-4}). This strategy helps to stabilise the convergence of the multimodal architecture and prevents overfitting of the backbone.

To further accelerate convergence and avoid local minima, we employed the OneCycleLR learning rate scheduler. The learning rate for each parameter group was initially set to one-tenth of its maximum, gradually increased during the first 30% of training steps, and then annealed to a very low value following a cosine schedule for the remainder of training. The number of steps was determined by the number of batches per epoch and the total number of epochs.

Training was conducted with a batch size 16 for up to 30 epochs. Early stopping was implemented if the validation loss failed to improve for seven consecutive epochs. Gradient clipping with a maximum norm of 1.0 was applied throughout to enhance training stability.

Model performance during training and validation was monitored using several standard metrics. Root mean squared error (RMSE) was used as the primary loss and evaluation metric for the regression of expert attention scores, as shown in (1). Where N is the total number of samples, y_i denotes the ground-truth label (expert attention score) for the i-th sample, and $\hat{y}_i$ is the predicted score for the i-th sample.

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} (\hat{y}_i - y_i)^2} \tag{1}$$

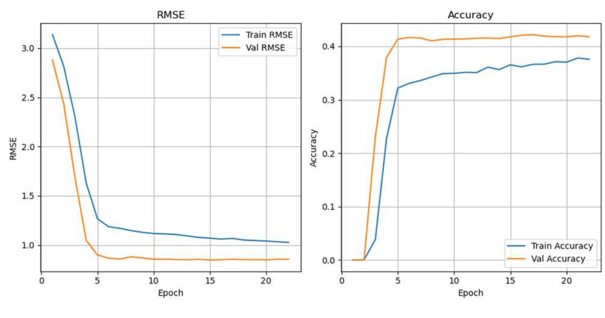
Additionally, classification accuracy was computed by rounding predicted attention scores to the nearest integer and comparing them to ground-truth labels, as shown in (2). For further analysis of attention state detection, binary accuracy, F1 score, and area under the ROC curve (AUC) were also calculated by binarizing the attention levels (with a threshold of 3). Where $\tilde{y}_i$denotes the tolerant rounded prediction for the i-th sample, τ is the tolerance threshold 0.15, y_i is the ground-truth label, and $\hat{y}_i$ is the predicted score.

$$\tilde{y}_i = \begin{cases} \text{Round}(\hat{y}_i), & |\hat{y}_i - \text{Round}(\hat{y}_i)| \le \tau \\ 3, & 2.85 < \hat{y}_i < 3.15 \\ \text{Round}(\hat{y}_i), & \text{οτηερωισε} \end{cases} \tag{2}$$

IV. Results and Discussion

A. Training Process Analysis

The training dynamics of the proposed multimodal model are illustrated in Fig. 3 and Fig. 4, which present the RMSE, classification accuracy, and attention loss curves for both the training and validation sets over 22 epochs. These figures show that the model demonstrates rapid convergence, with RMSE and accuracy stabilising after approximately 10 epochs. The validation RMSE drops sharply from 2.88 at the first epoch to below 1.0 within the first five epochs, and continues to decrease gradually, reaching its minimum value of 0.85 by epoch 15. Meanwhile, validation accuracy improves steadily from near zero to over 41% as the model learns to map the input features to attention scores.



(a) (b)

Fig. 3. Training and validation curves over epochs. (a) RMSE. (b) Accuracy.

In addition to RMSE and accuracy, the evolution of attention loss during training is presented in Figure 4. The training and validation attention loss curves exhibit a rapid and smooth decline within the first few epochs, with the validation loss decreasing from 8.30 in the initial epoch to approximately 0.73 at convergence. This significant reduction demonstrates the model's ability to accurately regress expert attention scores. The minimal gap between training and validation attention loss further indicates that the model generalises well and does not suffer from overfitting. These results highlight the effectiveness of our multimodal fusion framework and loss design in capturing attention states.

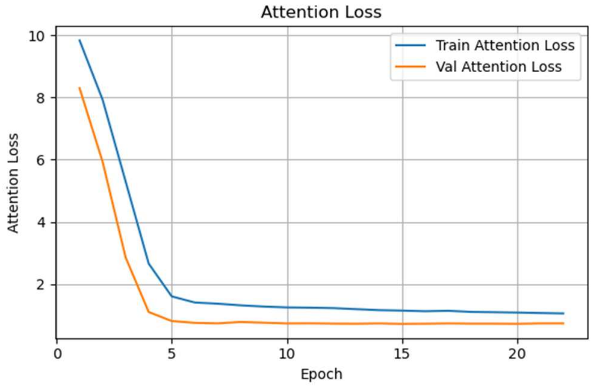


Fig. 4. Training and validation attention loss curves over epochs.

Early stopping was triggered after 22 epochs due to the lack of improvement in validation RMSE, demonstrating the stability and robustness of the training process. The minimal gap between training and validation curves indicates a low risk of overfitting and good generalisation capability of the model. Furthermore, the consistent improvement in accuracy and reduction in RMSE and attention loss across epochs highlights the effectiveness of the multimodal fusion approach and the model's ability to learn meaningful representations from diverse input signals.

B. Overall Performance Analysis

Table I summarises the best result of our proposed model on the hold-out test set, evaluated using several complementary metrics. The results demonstrate that the model achieves robust and reliable predictions of driver attention, both in continuous and discretised settings.

TABLE I. PERFORMANCE OF THE PROPOSED MODEL.

Metrics	**Results**
RMSE	0.8696
Tolerant Accuracy (±0.15)	0.4117
One-off Accuracy (±1)	0.7627
Binary Accuracy	0.6045
Binary F1 Score	0.7471
Binary AUC	0.5351

Among all the metrics, the RMSE and the one-off accuracy (±1) stand out as the most remarkable indicators of the model's performance. The RMSE of 0.87 is notably low, reflecting the model's exceptional ability to regress continuous expert attention scores with high precision. Such a low error demonstrates that our multimodal fusion and training strategies enable the model to capture subtle and nuanced variations in student attention.

The one-off accuracy (±1) reaches an impressive 76.3%, indicating that more than three-quarters of the model's predictions fall within one point of the ground truth label. This high value highlights the model's reliability and its robustness to minor fluctuations in expert scores, which are often unavoidable due to subjective annotation. The outstanding performance on one-off accuracy underscores the practical value of our approach for applications where precise but flexible attention estimation is required.

The tolerant accuracy (±0.15) reaches 41.2%, which is relatively modest compared to our other metrics. This result, however, is largely attributable to the edge effect inherent in the annotation and evaluation process: many ground-truth attention labels are near the boundaries of the rating scale, where even small prediction errors can easily exceed the narrow tolerance threshold, leading to a lower tolerance accuracy. Furthermore, the subjective nature of expert annotations can introduce additional variability, making perfect agreement within such a tight margin inherently challenging.

In the binary classification scenario (using a threshold of 3), the model achieves a binary accuracy of 60.5% and an F1 score of 0.75, confirming a strong balance between precision and recall for detecting high versus low attention states. While the binary AUC of 0.54 is moderate, it still indicates the model's ability to distinguish between attention classes above random chance, despite the inherent ambiguity and class imbalance in the data.

To sum up, these results validate the effectiveness and generalizability of our proposed approach. The combination of exceptionally low RMSE and high one-off accuracy demonstrates that our model not only predicts attention levels accurately but also provides robust and practically useful estimates. The consistent results across multiple metrics further reinforce the soundness of our model design, multimodal fusion strategy, and loss formulation, laying a strong foundation for subsequent deployment in real-world student attention monitoring systems.

V. CONCLUSION

A multimodal framework was proposed to automatically detect individual student attention in real-world classroom environments. By fusing synchronised facial video and wearable sensor data, our deep learning-based model leverages visual and physiological cues to provide a more accurate and robust estimation of student attention levels than single-modality approaches. Extensive experiments on the DIPSEER dataset demonstrate that our method achieves a notably low RMSE and high one-off accuracy, confirming its effectiveness in modelling subtle variations in attention. Although tolerant accuracy remains challenging due to the edge effect and inherent subjectivity of expert annotations, our model consistently delivers reliable and practical predictions across multiple evaluation metrics. Overall, our approach offers a solid foundation for objective and scalable student attention monitoring in face-to-face classroom settings.

Several future research directions are suggested, including (i) the adoption of data generation to enhance the data distribution and synthesize additional wearable data [19]; (ii) conducting research in image enhancement algorithm to improve the quality of facial images [20]; (iii) investigation of advanced noise filtering techniques [21]; and (iv) the exploration of cutting-edge artificial intelligence algorithms on classification problems [22].

REFERENCES

[1] M. B. Kaushik, B. M. Doshi, V. Gehlot, and S. K. Ranjan, “Unravelling the attention crisis: Investigating student engagement in contemporary classrooms,” Evergreen, vol. 11, no. 2, pp. 586–601, June 2024.

[2] M. V. Adler, J. Madsen, J. Hedberg, R. Steinberg, and L. C. Parra, “Effect of explanation videos on learning: The role of attention and academic performance,” Educ. Inf. Technol., vol. 30, pp. 1–29, January 2025.

[3] K. T. Chui, D. C. L. Fung, M. D. Lytras, and T. M. Lam, “Predicting at-risk university students in a virtual learning environment via a machine learning algorithm,” Comput. Human Behav., vol. 107, article no. 105584, June 2020.

[4] R. C. Pianta and B. K. Hamre, “Conceptualization, measurement, and improvement of classroom processes: Standardized observation can leverage capacity,” Educ. Res., vol. 38, no. 2, pp. 109–119, March 2009.

[5] O. Bulut et al., “The rise of artificial intelligence in educational measurement: Opportunities and ethical challenges,” 2024, arXiv:2406.18900. [Online]. Available: https://arxiv.org/abs/2406.18900

[6] X. M. Zhao, F. D. B. Yusop, H. C. Liu, Y. N. Prilanita, and Y. X. Chang, “Classroom student behavior recognition using an intelligent sensing framework,” IEEE Access, vol. 13, pp. 49767–49776, March 2025.

[7] P. Buono, B. De Carolis, F. D’Errico, N. Macchiarulo, and G. Palestra, “Assessing student engagement from facial behavior in on-line learning,” Multimed. Tools Appl., vol. 82, no. 9, pp. 12859–12877, April 2023.

[8] B. Alnasyan, M. Basheri, and M. Alassafi, “The power of deep learning techniques for predicting student performance in virtual learning environments: A systematic literature review,” Comput. Educ.: Artif. Intell., vol. 6, 100231, June 2024.

[9] Q. Liu, X. Jiang, and R. Jiang, “Classroom behavior recognition using computer vision: A systematic review,” Sensors, vol. 25, no. 2, 373, January 2025.

[10] H. Wang et al., “Automated student classroom behaviors’ perception and identification using motion sensors,” Bioengineering, vol. 10, no. 2, 127, January 2023.

[11] E. Dimitriadou and A. Lanitis, “A critical evaluation, challenges, and future perspectives of using artificial intelligence and emerging technologies in smart classrooms,” Smart Learning Environments, vol. 10, no. 12, February 2023.

[12] S. Wu et al., “A comprehensive exploration of personalized learning in smart education: From student modeling to personalized recommendations,” 2024, arXiv:2402.01666. [Online]. Available: https://arxiv.org/abs/2402.01666

[13] Y. Qi, L. Zhuang, H. Chen, X. Han, and A. Liang, “Evaluation of students’ learning engagement in online classes based on multimodal vision perspective,” Electronics, vol. 13, no. 1, 149, December 2023.

[14] Z. Trabelsi, F. Alnajjar, M. M. M. A. Parambil, M. Gochoo, and L. Ali, “Real-time attention monitoring system for classroom: A deep learning approach for student’s behavior recognition,” Big Data Cogn. Comput., vol. 7, no. 1, 48, March 2023.

[15] Y. Wang, “Multimodal data-supported learning engagement analysis,” Technology, Pedagogy and Education, vol. 34, no. 1, February 2025.

[16] Z. Wang, M. Wang, C. Zeng, and L. Li, “Multi-scale deformable transformers for student learning behavior detection in smart classroom,” 2024, arXiv:2410.07834. [Online]. Available: https://arxiv.org/abs/2410.07834

[17] L. Marquez-Carpintero et al., “DIPSER: A dataset for in-person student engagement recognition in the wild,” 2025, arXiv:2502.20209. [Online]. Available: https://arxiv.org/abs/2502.20209

[18] M. Tan and Q. V. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” in Proc. Int. Conf. Mach. Learn., 2020, pp. 6105–6114. [Online]. Available: https://proceedings.mlr.press/v97/tan19a.html

[19] J. T. W. Chan, K. T. Chui, L. K. Lee, N. Paoprasert, and K. K. Ng, “Data generation using a probabilistic auto-regressive model with application to student exam performance analysis,” in 2024 International Symposium on Educational Technology, Macao SAR, China, 29 July 2024 – 1 August 2024, pp. 87–90.

[20] R. W. Liu, Y. Guo, Y. Lu, K. T. Chui, and B. B. Gupta, “Deep network-enabled haze visibility enhancement for visual IoT-driven intelligent transportation systems,” IEEE Trans. Ind. Inform., vol. 19, no. 2, pp. 1581–1591, Feb. 2023

[21] N. K. Yadav, A. Dhawan, M. Tiwari, and S. K. Jha, “A state-of-the-art survey on noise removal in a non-stationary signal using adaptive finite impulse response filtering: challenges, techniques, and applications,” Int. J. Syst. Sci., vol. 56, no. 4, pp. 885–918, 2025.

[22] K. T. Chui, L. K. Lee, F. L. Wang, S. K. Cheung, and L. P. Wong, “A Review of Data Augmentation and Data Generation Using Artificial Intelligence in Education,” in International Conference on Technology in Education, Bangkok, Thailand, 19–21 Dec. 2023, pp. 242–253.